Importing Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
import plotly.express as px
```

Data Loading

```
df = pd.read_csv("C:/Users/Admin/Desktop/Capstone/pizza_sales.csv")

df.head()
```

```
   pizza_id  order_id  pizza_name_id  quantity  order_date order_time
\
0         1         1     hawaiian_m         1  01-01-2015   11:38:36

1         2         2  classic_dlx_m         1  01-01-2015   11:57:40

2         3         2  five_cheese_l         1  01-01-2015   11:57:40

3         4         2    ital_supr_l         1  01-01-2015   11:57:40

4         5         2     mexicana_m         1  01-01-2015   11:57:40


   unit_price  total_price pizza_size pizza_category  \
0       13.25        13.25          M        Classic
1       16.00        16.00          M        Classic
2       18.50        18.50          L         Veggie
3       20.75        20.75          L        Supreme
4       16.00        16.00          M         Veggie

                                   pizza_ingredients  \
0           Sliced Ham, Pineapple, Mozzarella Cheese
1  Pepperoni, Mushrooms, Red Onions, Red Peppers,...
2  Mozzarella Cheese, Provolone Cheese, Smoked Go...
3  Calabrese Salami, Capocollo, Tomatoes, Red Oni...
4  Tomatoes, Red Peppers, Jalapeno Peppers, Red O...

                  pizza_name
0         The Hawaiian Pizza
1   The Classic Deluxe Pizza
2      The Five Cheese Pizza
```

```
3  The Italian Supreme Pizza
4         The Mexicana Pizza
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 48620 entries, 0 to 48619
Data columns (total 12 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   pizza_id           48620 non-null  int64
 1   order_id           48620 non-null  int64
 2   pizza_name_id      48620 non-null  object
 3   quantity           48620 non-null  int64
 4   order_date         48620 non-null  object
 5   order_time         48620 non-null  object
 6   unit_price         48620 non-null  float64
 7   total_price        48620 non-null  float64
 8   pizza_size         48620 non-null  object
 9   pizza_category     48620 non-null  object
 10  pizza_ingredients  48620 non-null  object
 11  pizza_name         48620 non-null  object
dtypes: float64(2), int64(3), object(7)
memory usage: 4.5+ MB
```

```
df.describe()
```

```
           pizza_id      order_id      quantity    unit_price
total_price
count  48620.000000  48620.000000  48620.000000  48620.000000
48620.000000
mean   24310.500000  10701.479761      1.019622     16.494132
16.821474
std    14035.529381   6180.119770      0.143077      3.621789
4.437398
min        1.000000      1.000000      1.000000      9.750000
9.750000
25%    12155.750000   5337.000000      1.000000     12.750000
12.750000
50%    24310.500000  10682.500000      1.000000     16.500000
16.500000
75%    36465.250000  16100.000000      1.000000     20.250000
20.500000
max    48620.000000  21350.000000      4.000000     35.950000
83.000000
```

KPI'S

```
total_revenue = df['total_price'].sum()
total_pizzas_sold = df['quantity'].sum()
total_orders = df['order_id'].nunique()
avg_order_value = total_revenue / total_orders
avg_pizzas_per_order = total_pizzas_sold / total_orders

print(f"Total Revenue: ${total_revenue:,.2f}")
print(f"Total Pizzas Sold: {total_pizzas_sold:,}")
print(f"Total Orders: {total_orders:,}")
print(f"Avg Order Value: ${avg_order_value:,.2f}")
print(f"Average Pizza per Order: {avg_pizzas_per_order:.2f}")
```

```
Total Revenue: $817,860.05
Total Pizzas Sold: 49,574
Total Orders: 21,350
Avg Order Value: $38.31
Average Pizza per Order: 2.32
```

CHART's

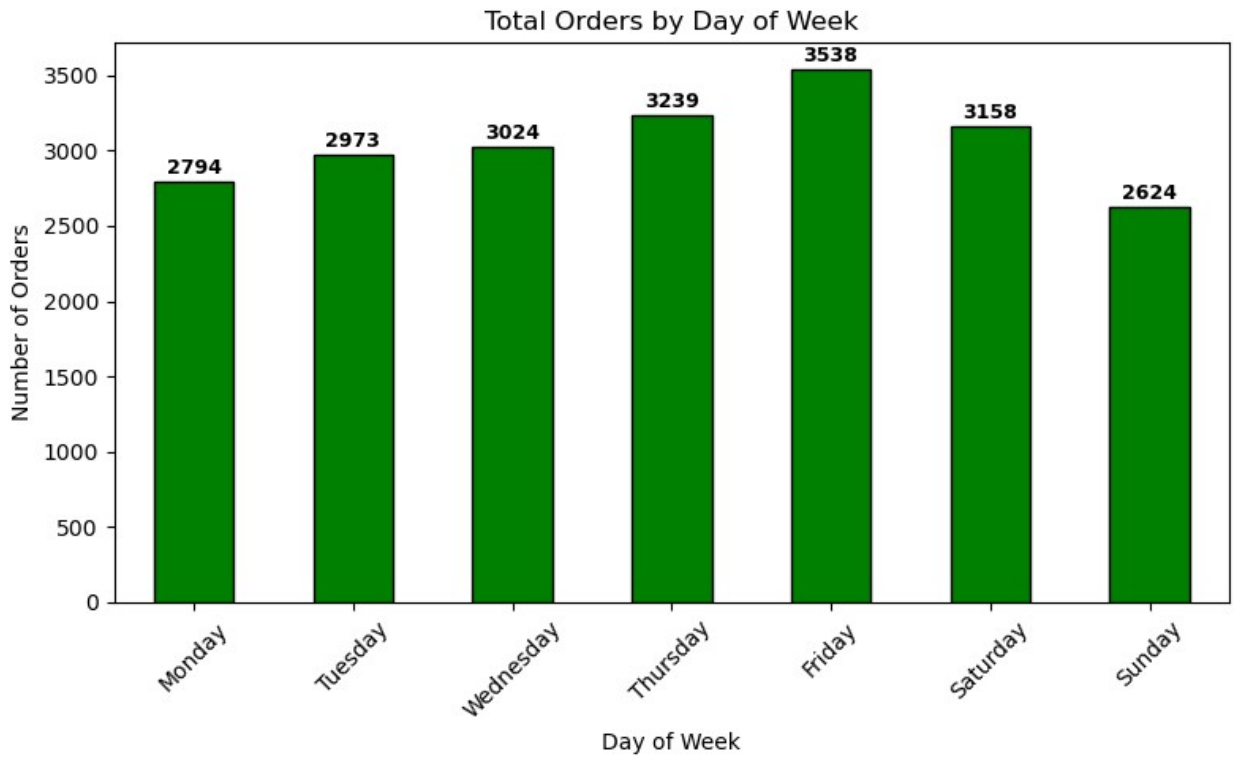
Ingredient Analysis

```
ingredient = (
    df['pizza_ingredients']
        .str.split(',')
        .explode()
        .str.strip()
        .value_counts()
        .reset_index()
        .rename(columns={'index': 'Indegredients',
'pizza_ingredients': 'counts'})
)

print(ingredient.head(10))
```

```
               counts  count
0              Garlic  27422
1            Tomatoes  26601
2          Red Onions  19547
3         Red Peppers  16284
4  Mozzarella Cheese  10333
5           Pepperoni  10300
6             Spinach  10012
7           Mushrooms   9624
8             Chicken   8443
9           Capocollo   6572
```

1.Daily Trend – Total Orders

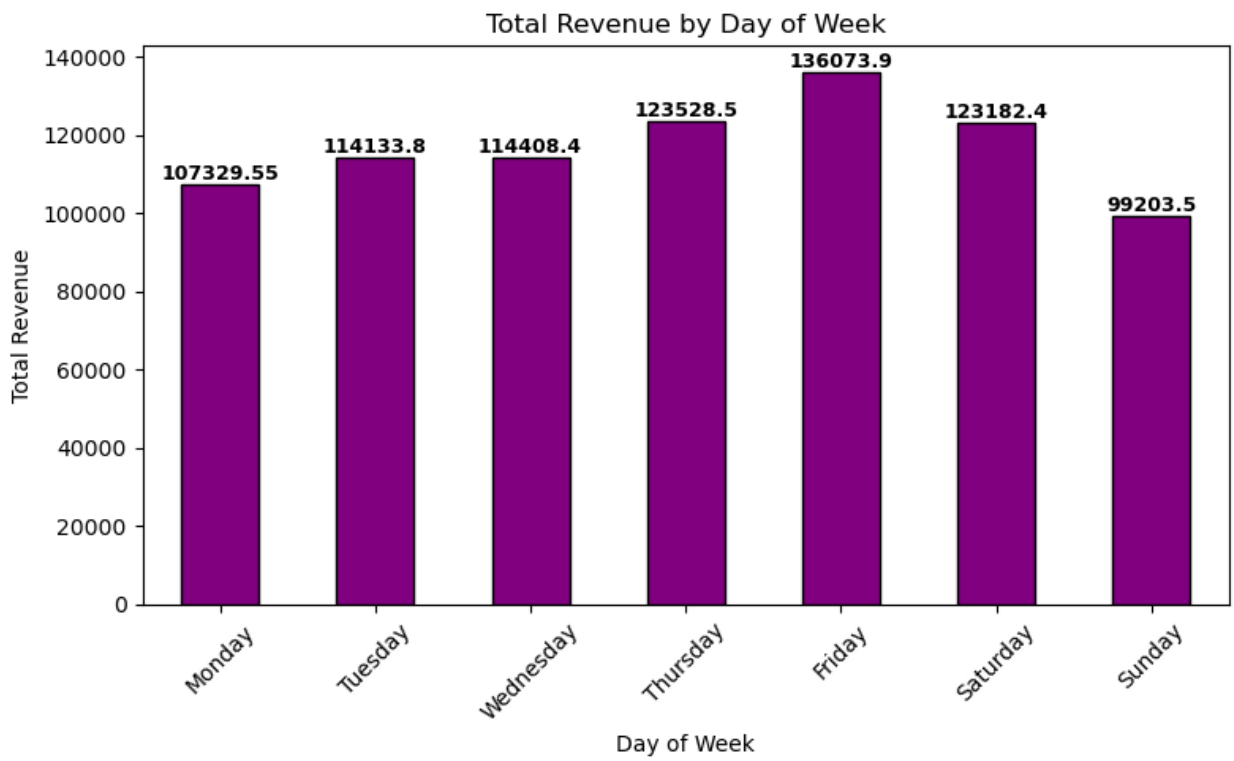
```
df['order_date'] = pd.to_datetime(df['order_date'], dayfirst=True)

df['day_name'] = df['order_date'].dt.day_name()

weekday_order = ["Monday", "Tuesday", "Wednesday", "Thursday",
"Friday", "Saturday", "Sunday"]

df['day_name'] = pd.Categorical(df['day_name'],
categories=weekday_order, ordered=True)

orders_by_day = df.groupby('day_name', observed=False)
['order_id'].nunique()

ax = orders_by_day.plot(kind='bar', figsize=(8,5), color='green',
edgecolor='black')

plt.title("Total Orders by Day of Week")
plt.xlabel("Day of Week")
plt.ylabel("Number of Orders")
plt.xticks(rotation=45)

for i, val in enumerate(orders_by_day):
    plt.text(i, val + 20, str(val), ha='center', va='bottom',
fontsize=9, fontweight='bold')

plt.tight_layout()
plt.show()
```



2.Daily Trend – Total Revenue

```
df['order_date'] = pd.to_datetime(df['order_date'], dayfirst=True)

df['day_name'] = df['order_date'].dt.day_name()

weekday_order = ["Monday", "Tuesday", "Wednesday", "Thursday",
"Friday", "Saturday", "Sunday"]

df['day_name'] = pd.Categorical(df['day_name'],
categories=weekday_order, ordered=True)

orders_by_day = df.groupby('day_name', observed=False)
['total_price'].sum()

ax = orders_by_day.plot(kind='bar', figsize=(8,5), color='purple',
edgecolor='black')

plt.title("Total Revenue by Day of Week")
plt.xlabel("Day of Week")
plt.ylabel("Total Revenue")
plt.xticks(rotation=45)

for i, val in enumerate(orders_by_day):
    plt.text(i, val + 20, str(val), ha='center', va='bottom',
```

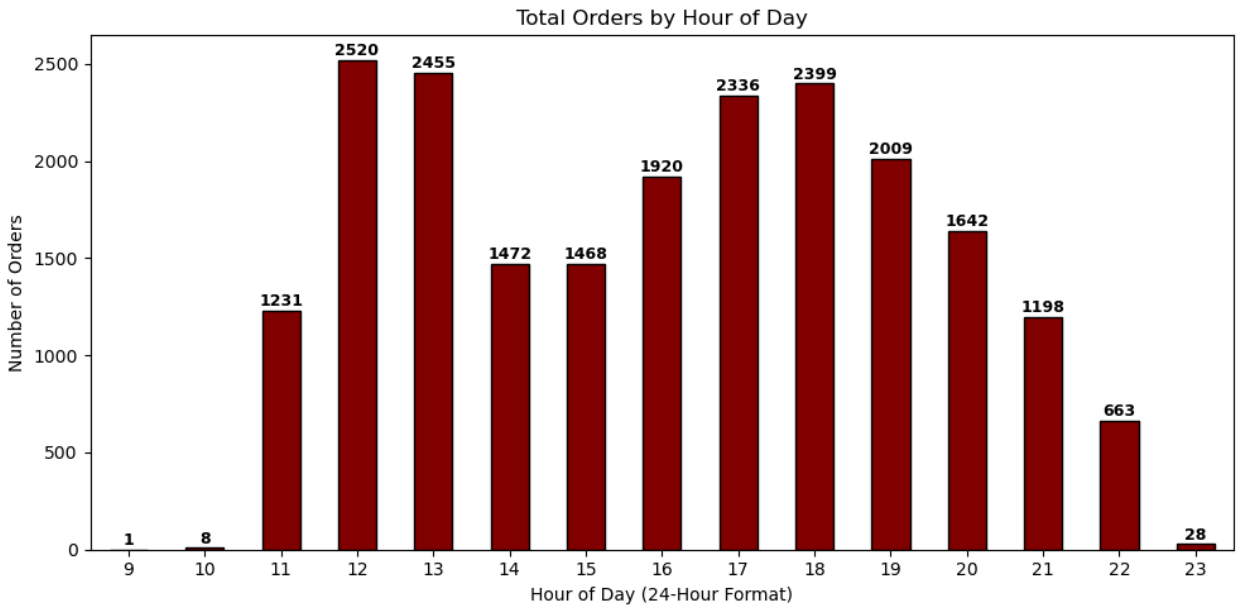
```
fontsize=9, fontweight='bold')

plt.tight_layout()
plt.show()
```



Hourly Trend-Total Orders

```
df['order_time'] = pd.to_datetime(df['order_time'], format='%H:%M:%S')

df['order_hour'] = df['order_time'].dt.hour

orders_by_hour = df.groupby('order_hour', observed=False)
['order_id'].nunique()

ax = orders_by_hour.plot(kind='bar', figsize=(10,5), color='maroon',
edgecolor='black')

plt.title("Total Orders by Hour of Day")
plt.xlabel("Hour of Day (24-Hour Format)")
plt.ylabel("Number of Orders")
plt.xticks(rotation=0)

for i, val in enumerate(orders_by_hour):
    plt.text(i, val + 5, str(val), ha='center', va='bottom',
fontsize=9, fontweight='bold')
```

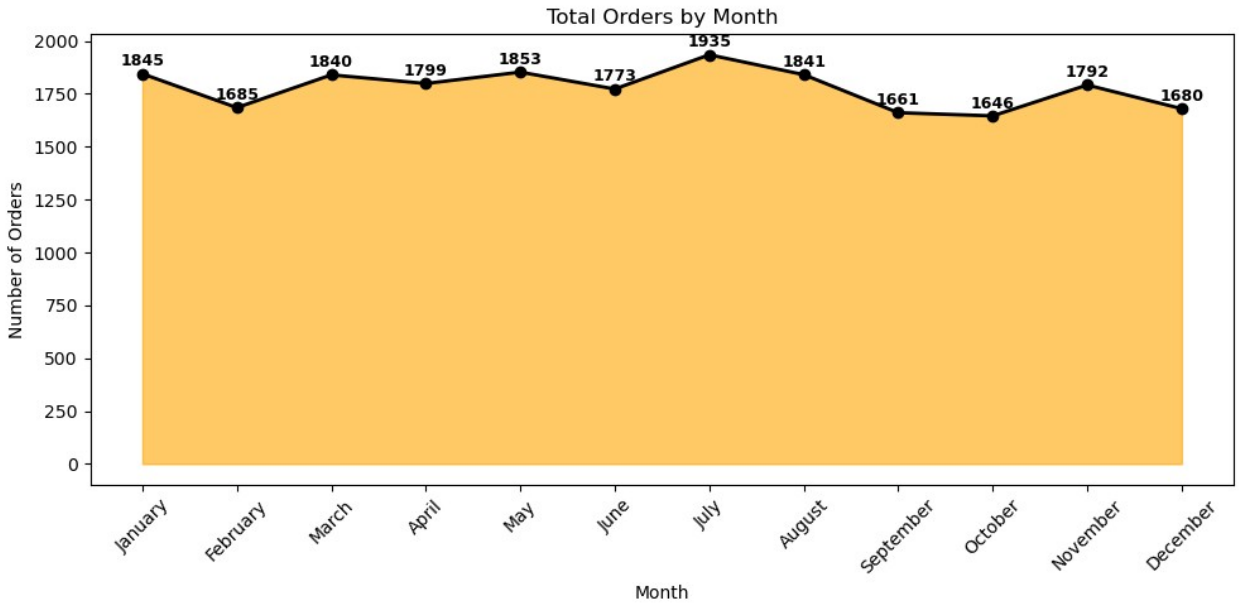
```
plt.tight_layout()
plt.show()
```



Monthly Trend – Total Orders

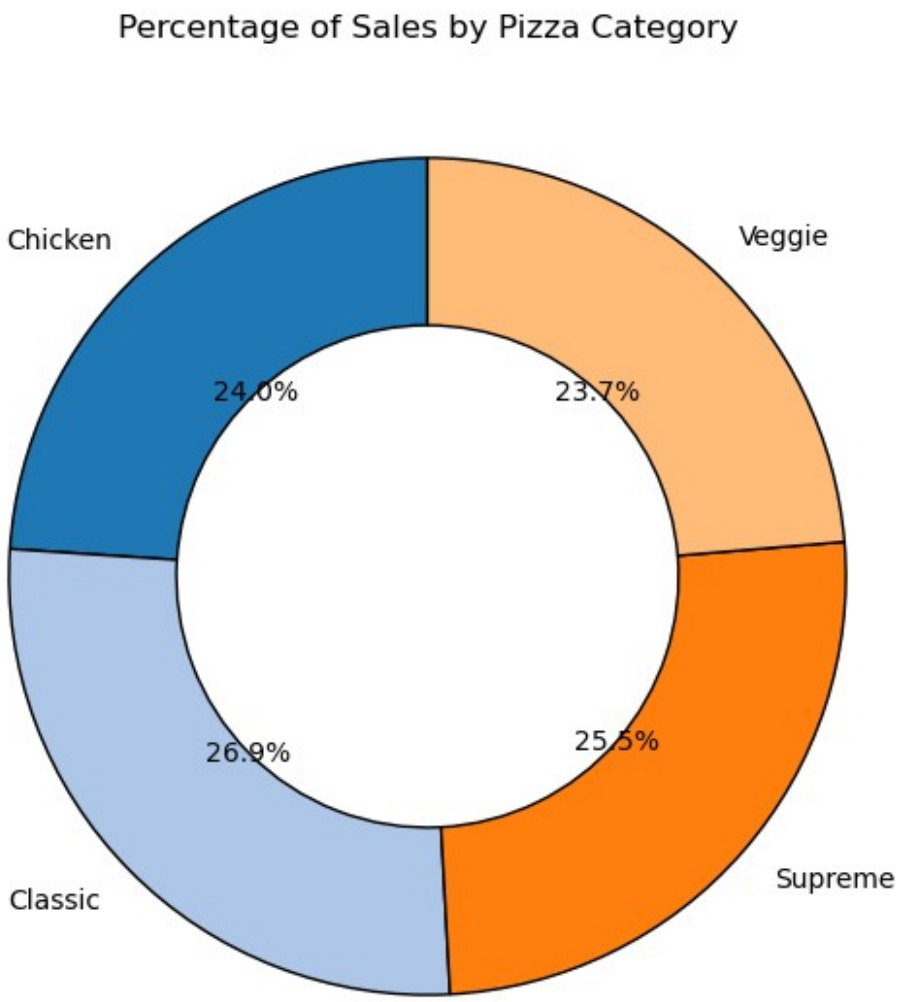
```
df['order_date'] = pd.to_datetime(df['order_date'], dayfirst=True)

df['month_name'] = df['order_date'].dt.month_name()

month_order = ["January", "February", "March", "April", "May", "June",
                "July", "August", "September", "October", "November",
"December"]

df['month_name'] = pd.Categorical(df['month_name'],
categories=month_order, ordered=True)

orders_by_month = df.groupby('month_name', observed=False)
['order_id'].nunique()

plt.figure(figsize=(10,5))
plt.fill_between(orders_by_month.index, orders_by_month.values,
color="orange", alpha=0.6)
plt.plot(orders_by_month.index, orders_by_month.values, color="black",
linewidth=2, marker='o')

plt.title("Total Orders by Month")
plt.xlabel("Month")
plt.ylabel("Number of Orders")
```

```
plt.xticks(rotation=45)

for i, val in enumerate(orders_by_month):
    plt.text(i, val + 20, str(val), ha='center', va='bottom',
fontsize=9, fontweight='bold')

plt.tight_layout()
plt.show()
```



% of Sales by Category

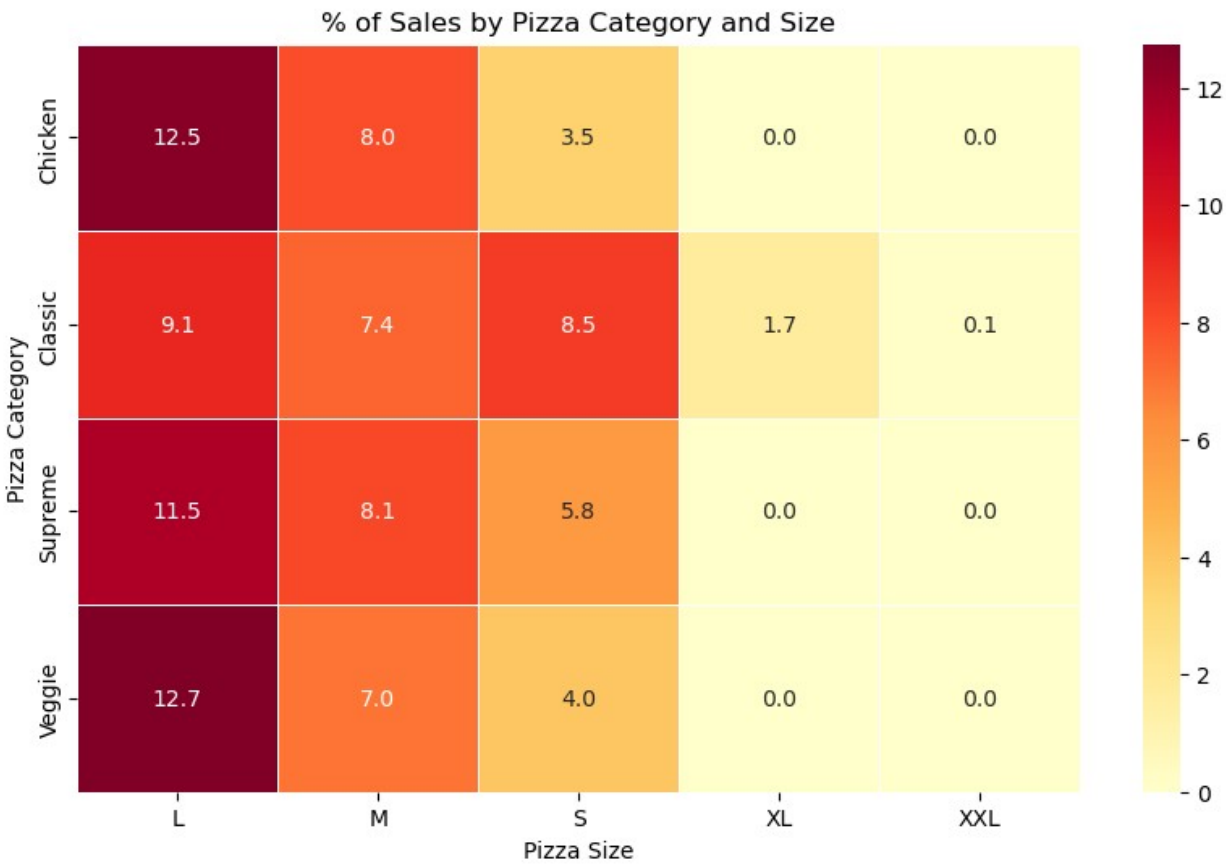
```
category_sales = df.groupby('pizza_category')['total_price'].sum()

category_pct = category_sales / category_sales.sum() * 100

plt.figure(figsize=(7,7))
colors = plt.get_cmap('tab20').colors  # nice color palette

plt.pie(category_pct, labels=category_pct.index, autopct='%1.1f%%',
        startangle=90, colors=colors,
        wedgeprops={'edgecolor': 'black', 'width': 0.4})

plt.title("Percentage of Sales by Pizza Category")
plt.show()
```



% Sales by Pizza Size & Category

```
sales_pivot = df.pivot_table(
    index='pizza_category',
    columns='pizza_size',
    values='total_price',
    aggfunc='sum',
    fill_value=0
)

sales_pct = sales_pivot / sales_pivot.sum().sum() * 100

plt.figure(figsize=(10,6))
sns.heatmap(sales_pct, annot=True, fmt=".1f", cmap="YlOrRd",
linewidths=0.5)
```

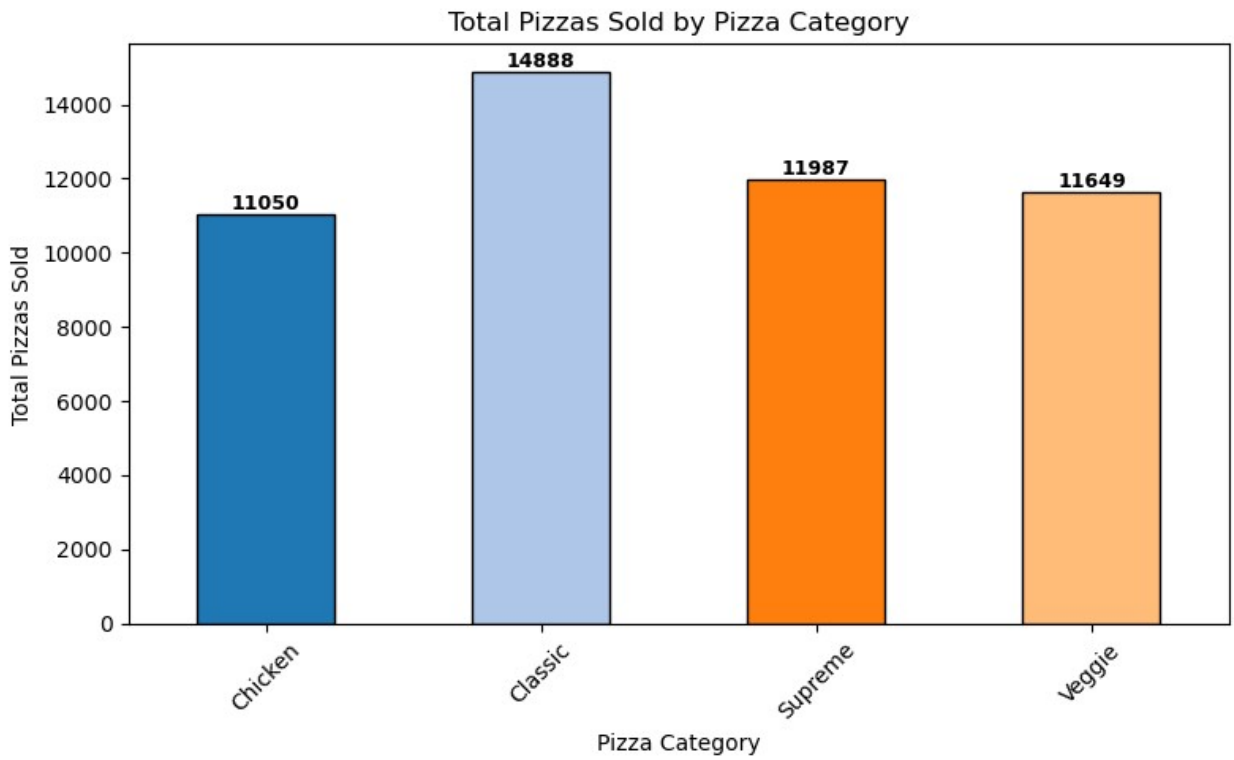
```
plt.title("% of Sales by Pizza Category and Size")
plt.ylabel("Pizza Category")
plt.xlabel("Pizza Size")

plt.show()
```



Total Pizzas Sold by Pizza Category

```
pizzas_by_category = df.groupby('pizza_category')['quantity'].sum()

colors = list(plt.get_cmap('tab20').colors)
colors = colors[:len(pizzas_by_category)]

ax = pizzas_by_category.plot(kind='bar', figsize=(8,5), color=colors,
edgecolor='black')

plt.title("Total Pizzas Sold by Pizza Category")
plt.xlabel("Pizza Category")
plt.ylabel("Total Pizzas Sold")
plt.xticks(rotation=45)
```

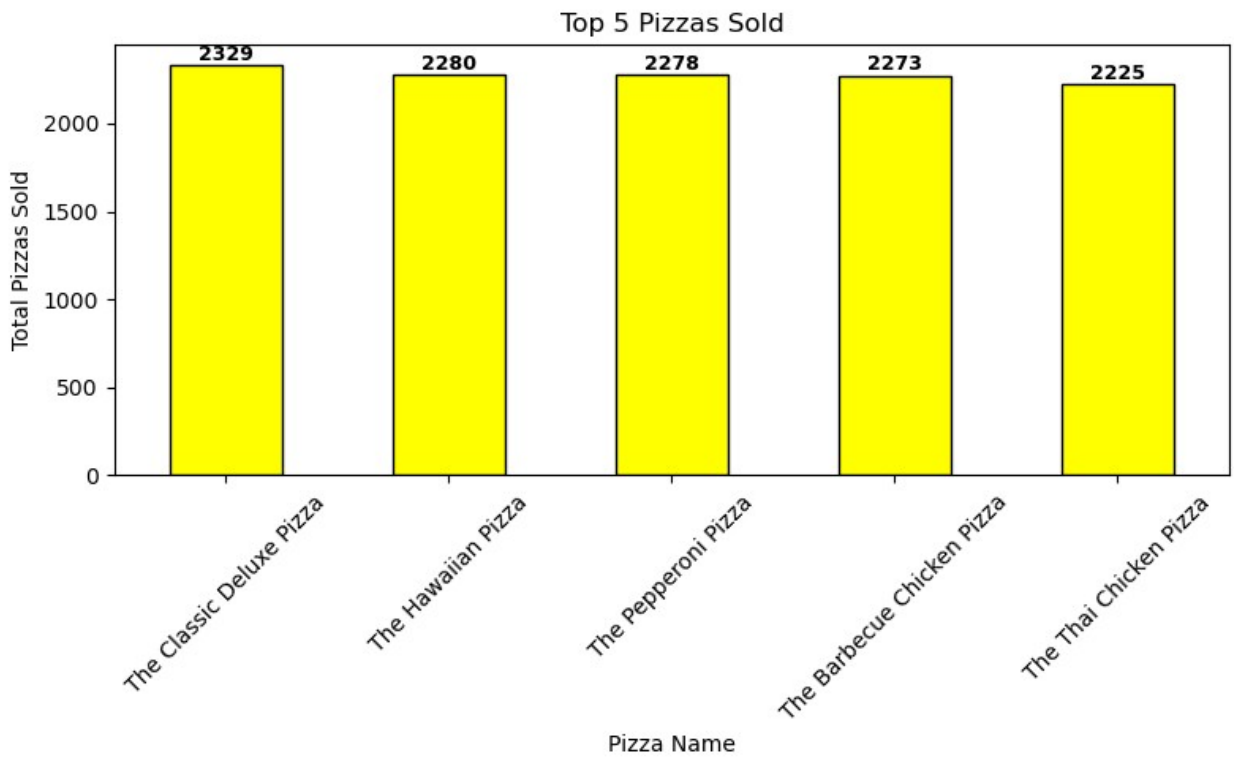
```
for i, val in enumerate(pizzas_by_category):
    plt.text(i, val + 5, str(val), ha='center', va='bottom',
fontsize=9, fontweight='bold')

plt.tight_layout()
plt.show()
```



Top 5 Best-Selling Pizzas – Total Orders

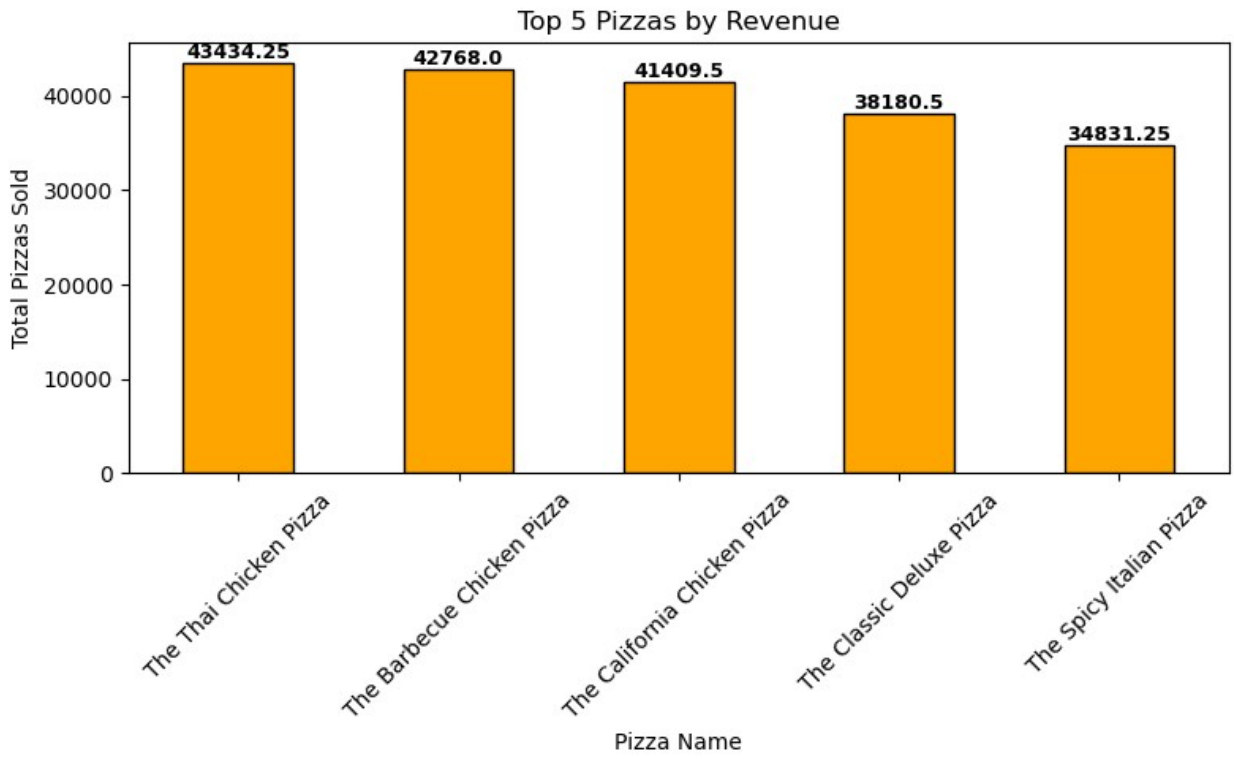
```
pizzas_by_name = df.groupby('pizza_name')['order_id'].nunique()

top5 = pizzas_by_name.sort_values(ascending=False).head(5)

ax = top5.plot(kind='bar', figsize=(8,5), color='yellow',
edgecolor='black')

plt.title("Top 5 Pizzas Sold")
plt.xlabel("Pizza Name")
plt.ylabel("Total Pizzas Sold")
plt.xticks(rotation=45)

for i, val in enumerate(top5):
    plt.text(i, val + 2, str(val), ha='center', va='bottom',
fontsize=9, fontweight='bold')
```

```
plt.tight_layout()
plt.show()
```



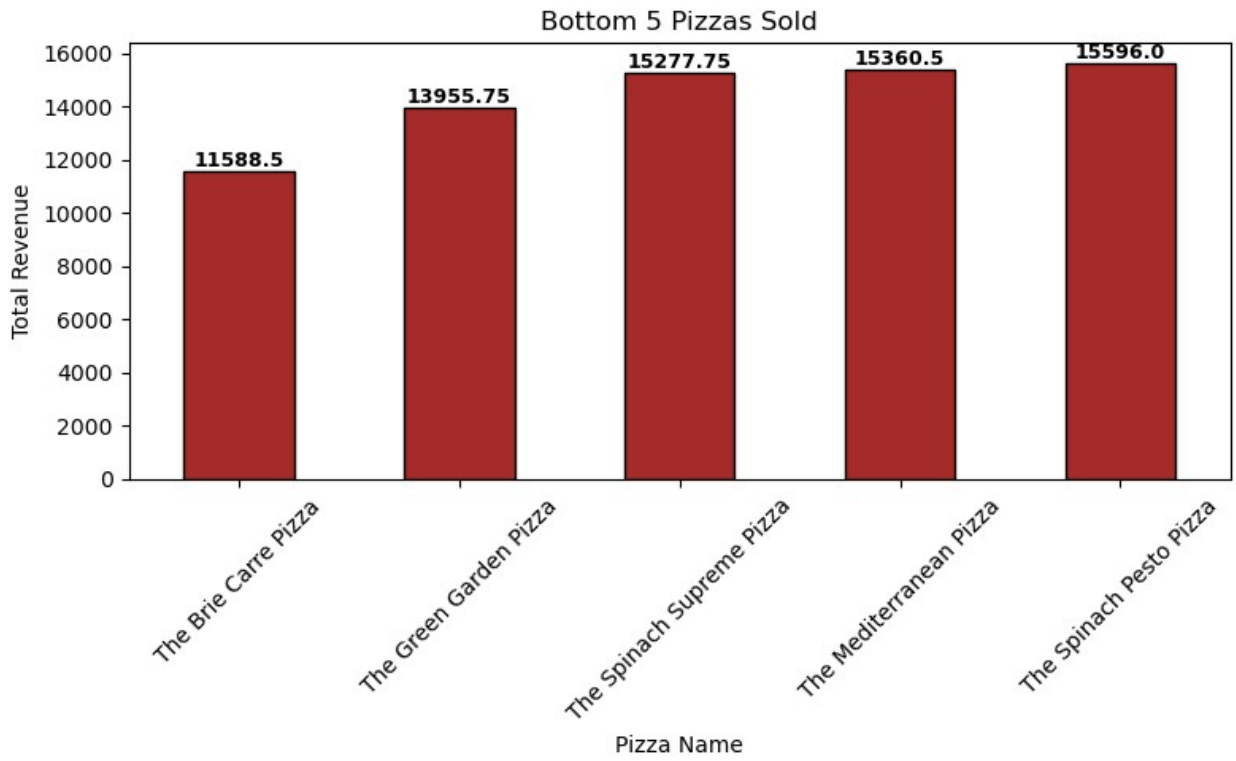
Top 5 Best-Selling Pizzas – Total Sales/Revenue

```
pizzas_by_name = df.groupby('pizza_name')['total_price'].sum()

top5 = pizzas_by_name.sort_values(ascending=False).head(5)

ax = top5.plot(kind='bar', figsize=(8,5), color='orange',
edgecolor='black')

plt.title("Top 5 Pizzas by Revenue")
plt.xlabel("Pizza Name")
plt.ylabel("Total Pizzas Sold")
plt.xticks(rotation=45)

for i, val in enumerate(top5):
    plt.text(i, val + 2, str(val), ha='center', va='bottom',
fontsize=9, fontweight='bold')

plt.tight_layout()
plt.show()
```



Bottom 5 Least-Selling Pizzas-Total Sales

```
pizzas_by_name = df.groupby('pizza_name')['total_price'].sum()
bottom5 = pizzas_by_name.sort_values(ascending=True).head(5)

ax = bottom5.plot(kind='bar', figsize=(8,5), color='brown',
edgecolor='black')
plt.title("Bottom 5 Pizzas Sold")
plt.xlabel("Pizza Name")
plt.ylabel("Total Revenue")
plt.xticks(rotation=45)

for i, val in enumerate(bottom5):
    plt.text(i, val + 2, str(val), ha='center', va='bottom',
fontsize=9, fontweight='bold')

plt.tight_layout()
plt.show()
```



Business Questions Answered

1.What is the total revenue generated?

```
total_revenue = df['total_price'].sum()
total_revenue
```

```
np.float64(817860.0499999999)
```

```
total_revenue
```

```
np.float64(817860.0499999999)
```

Total Revenue Generated = $817,860.05

2.How many pizzas were sold in total?

```
total_pizzas_sold = df['quantity'].sum()
total_pizzas_sold
```

```
np.int64(49574)
```

3.Which category and size of pizzas perform best?

```
# Category performance:
df.groupby('pizza_category')['total_price'].sum()

pizza_category
Chicken    195919.50
Classic    220053.10
Supreme    208197.00
Veggie     193690.45
Name: total_price, dtype: float64

# Size performance:
df.groupby('pizza_size')['total_price'].sum()

pizza_size
L      375318.70
M      249382.25
S      178076.50
XL      14076.00
XXL      1006.60
Name: total_price, dtype: float64
```

4.Which pizzas are the top and bottom performers?

```
# Top selling pizzas
df.groupby('pizza_name')
['quantity'].sum().sort_values(ascending=False).head(5)

pizza_name
The Classic Deluxe Pizza      2453
The Barbecue Chicken Pizza    2432
The Hawaiian Pizza            2422
The Pepperoni Pizza           2418
The Thai Chicken Pizza        2371
Name: quantity, dtype: int64
```

```
# Lowest selling pizzas
df.groupby('pizza_name')['quantity'].sum().sort_values().head(5)
```

```
pizza_name
The Brie Carre Pizza         490
The Mediterranean Pizza      934
The Calabrese Pizza          937
The Spinach Supreme Pizza    950
The Soppressata Pizza        961
Name: quantity, dtype: int64
```

5.What is the average order value & average pizzas per order?

```
avg_order_value = total_revenue / df['order_id'].nunique()
avg_order_value
```

```
np.float64(38.30726229508196)
```

```
avg_pizzas_per_order = df['quantity'].sum() / df['order_id'].nunique()
avg_pizzas_per_order
```

```
np.float64(2.321967213114754)
```

What are the sales trends by day, month, and time of day?

Daily trend

```
df.groupby(df['order_date'].dt.day_name())['order_id'].nunique()
```

```
order_date
Friday       3538
Monday       2794
Saturday     3158
Sunday       2624
Thursday     3239
Tuesday      2973
Wednesday    3024
Name: order_id, dtype: int64
```

Hourly trend

```
df.groupby(df['order_time'].dt.hour)['order_id'].nunique()
```

```
order_time
9        1
10       8
11    1231
12    2520
13    2455
14    1472
15    1468
16    1920
17    2336
18    2399
19    2009
20    1642
21    1198
22     663
23      28
Name: order_id, dtype: int64
```

Monthly trend

```
df.groupby(df['order_date'].dt.month_name())['order_id'].nunique()
```

```
order_date
April        1799
August       1841
December     1680
February     1685
January      1845
July         1935
June         1773
March        1840
May          1853
November     1792
October      1646
September    1661
Name: order_id, dtype: int64
```